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Sáb 16/05/2026 1:08 AM

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Hello,

The following submission has been created.

Track Name: LoHacemosxTIC2026

Paper ID: 18

Paper Title: Quantitative Precipitation Estimation Model Based on Data from Different C-Band Radars in Colombia

Abstract:

This research focuses on the development and validation of a proposed Quantitative Precipitation Estimation (QPE) model designed to improve the accuracy of rainfall quantification. The model uses data from two radars with overlapping coverage in Colombia. Data from the Carimagua and Barrancabermeja radars are used for this study, integrating polarimetric parameters with ground-based rainfall data from surface rain gauge stations. The methodological development begins with estimating rainfall from radar data using the conventional methods $R(Z)$, $R(KDP)$, and $R(Z, ZDR)$. Following the analysis of the initial results, conducted using error metrics (MAE, RMSE, MAPE, PBIAS, WMAPE, and NRMSE), it was determined that the reflectivity ratio $R(Z)$ is the most appropriate among the conventional methods applied.

Based on this, the project focuses on designing a quantitative precipitation estimation (QPE) model based on Artificial Neural Networks (ANNs). This model independently processes the parameters of each radar to ultimately arrive at a unified neural network, which achieves a reduction in precipitation estimation error compared to traditional models and individual networks. The results obtained represent an advance in the characterization of the national radar network by integrating data in overlapping coverage areas, thus improving QPE accuracy through a unified model.

Created on: Sat, 16 May 2026 06:08:08 GMT

Last Modified: Sat, 16 May 2026 06:08:08 GMT

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Secondary Subject Areas: Not Entered

Submission Files: Not Uploaded

Submission Questions Response:

1. Topics
Technology and artificial intelligence
2. Additional Information
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3. I am interested in having the work submitted for publication in a scientific journal
Agreement accepted
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